JAVA MULTILEVEL INHERITANCE PROGRAM

Object-Oriented Programming Practical Record

1. Aim

To write a Java program using multilevel inheritance to implement a student result system that calculates total marks, average and grade.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named Main.java.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create an object for Grade.
6. Grade inherits the properties and methods of Result.
7. Result inherits the properties and methods of Student.
8. Call the total, average and grade methods.
9. Display the student result.

3. Algorithm

1. Start the program.
2. Create the parent class Student.
3. Declare three integer variables m1, m2 and m3.
4. Create total() method in the Student class.
5. Create the Result class extending Student.
6. Create calculateAverage() method in the Result class.
7. Create the Grade class extending Result.
8. Create calculateGrade() method in the Grade class.
9. Create an object of Grade in main().
10. Call total(), calculateAverage() and calculateGrade().
11. Display the results.
12. Stop the program.

4. Explanation of the Program

Student class: It is the parent class containing three integer variables, m1, m2 and m3. It also contains the total() method.

Result class: It is the child class that extends Student. It inherits the marks and total() method and provides the calculateAverage() method.

Grade class: It extends Result. Therefore, it indirectly inherits the properties and methods of Student and provides the calculateGrade() method.

Multilevel Inheritance: The program demonstrates multilevel inheritance because Grade inherits from Result, and Result inherits from Student.

Main class: The main() method creates an object of Grade and calls all three result operations.

5. UML Class Diagram

```
+-----------------------------+
|           Student           |
+-----------------------------+
| - m1 : int = 80             |
| - m2 : int = 75             |
| - m3 : int = 90             |
+-----------------------------+
| + total() : void            |
+-------------+---------------+
              |
           extends
              v
+-----------------------------+
|           Result            |
+-----------------------------+
| - average : double          |
+-----------------------------+
| + calculateAverage() : void |
+-------------+---------------+
              |
           extends
              v
+-----------------------------+
|            Grade            |
+-----------------------------+
| + calculateGrade() : void   |
+-------------+---------------+
              ^
              |
        object created
              |
+-----------------------------+
|            Main             |
+-----------------------------+
| + main() : void             |
+-----------------------------+
```

6. Java Source Code

```
// Multilevel Inheritance
// Real application: Student Result

class Student {
    int m1 = 80;
    int m2 = 75;
    int m3 = 90;

    void total() {
        System.out.println("Total = " + (m1 + m2 + m3));
    }
}

class Result extends Student {

    double average;

    void calculateAverage() {
        average = (m1 + m2 + m3) / 3.0;
        System.out.println("Average = " + average);
    }
}

class Grade extends Result {

    void calculateGrade() {
        if (average >= 80)
            System.out.println("Grade = A");
        else if (average >= 60)
            System.out.println("Grade = B");
        else
            System.out.println("Grade = C");
    }
}

public class Main {
    public static void main(String[] args) {

        Grade obj = new Grade();

        obj.total();
        obj.calculateAverage();
        obj.calculateGrade();
    }
}
```

7. Output of the Program

```
Total = 245
Average = 81.66666666666667
Grade = A
```

8. Result

Thus, the Java program successfully demonstrates multilevel inheritance using a real-world student result application. The Grade class inherits from Result, and Result inherits from Student. The program calculates total marks, average and grade.